

Exp 2 : Implement a TCP/IP socket-based Automated Teller Machine (ATM) System

- Server maintains **customer details** (card number, PIN, balance)
- Client sends **card number, PIN, withdrawal amount**
- Server:
 - Validates login
 - Displays welcome message
 - Performs withdrawal if balance is sufficient
 - Otherwise sends warning message

Server Program

1. Import required networking and I/O packages.
2. Create a **ServerSocket** with a specific port number.
3. Wait for client connection using **accept()**.
4. Establish **input and output data streams**.
5. Store customer details (card number, PIN, balance).
6. Receive card number, PIN, and withdrawal amount from client.
7. **Validate card number and PIN.**
8. Check for sufficient balance.
9. Process withdrawal or generate warning message.
10. **Send transaction status and remaining balance to client.**

Client Program

1. Import required networking and I/O packages.
2. Create a **socket connection to the server** using IP and port.
3. Establish **input and output data streams**.
4. Accept card number, PIN, and withdrawal amount from user.
5. **Send entered details to the server.**
6. Receive transaction status message from server.
7. Display remaining balance to the user.
8. **Close client socket** and streams.

- `import java.io.*;`
- `import java.net.*;`
- `import java.util.HashMap;`

```
// Customer class
class Customer {
    String name;
    String pin;
    double balance;

    Customer(String name, String pin, double balance) {
        this.name = name;
        this.pin = pin;
        this.balance = balance;
    }
}
```

```
public class ATMServer {  
    public static void main(String[] args) {  
        try {  
            ServerSocket serverSocket = new ServerSocket(6000);  
            System.out.println("ATM Server started on port 6000...");  
            Socket socket = serverSocket.accept();  
            System.out.println("Client connected.");  
            BufferedReader in = new BufferedReader(  
                new InputStreamReader(socket.getInputStream()));  
            PrintWriter out = new PrintWriter(socket.getOutputStream(),  
true);
```

```
// Customer database
```

```
HashMap<String, Customer> customers = new HashMap<>();  
customers.put("1111", new Customer("Arun Kumar", "1234", 15000));  
customers.put("2222", new Customer("Priya Sharma", "2222", 8700));  
customers.put("3333", new Customer("Rahul Mehta", "9999", 23000));  
while (true) {  
    // Authentication  
    out.println("Enter Card Number:");  
    String card = in.readLine();  
    out.println("Enter PIN:");  
    String pin = in.readLine();
```

```
out.println("AUTH_SUCCESS");
    Customer c = customers.get(card);
    out.println("Welcome " + c.name);
    boolean session = true;
    while (session) {
        out.println("MENU");
        out.println("1. Balance Enquiry");
        out.println("2. Withdraw");
        out.println("3. Exit");
        out.println("Enter Choice:");
    }
    int choice = Integer.parseInt(in.readLine());
```

```
switch (choice) {  
    case 1:  
        out.println("Your Balance: Rs. " + c.balance);  
        break;  
    case 2:  
        out.println("Enter amount to withdraw:");  
        double amt = Double.parseDouble(in.readLine());  
        if (amt <= c.balance) {  
            c.balance -= amt;  
            out.println("Withdrawal Successful!");  
            out.println("Remaining Balance: Rs. " + c.balance);  
        } else {  
            out.println("Insufficient Balance!");  
        }  
        break;  
}
```

```
case 3:
    out.println("Thank you for using ATM!");
    session = false;
    break;
default:
    out.println("Invalid Choice!");
}
}
}

} catch (Exception e) {
    e.printStackTrace();
}
}
}
```

```
import java.io.*;
import java.net.*;
import java.util.Scanner;
public class ATMClient {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 6000);
            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
            Scanner sc = new Scanner(System.in);
            while (true) {
                // Card number
                System.out.println(in.readLine());
                out.println(sc.nextLine());
            }
        }
    }
}
```

```
// PIN
    System.out.println(in.readLine());
    out.println(sc.nextLine());
    String status = in.readLine();
    if (status.equals("AUTH_FAIL")) {
        System.out.println("Invalid Card or PIN!\n");
        continue;
    }
    System.out.println("Authentication Successful");
    System.out.println(in.readLine());
    boolean session = true;
    while (session) {
        System.out.println(in.readLine());
        System.out.println(in.readLine());
        System.out.println(in.readLine());
        System.out.println(in.readLine());
    }
}
```

```
int choice = sc.nextInt();
    sc.nextLine();
    out.println(choice);
    if (choice == 1) {
        System.out.println(in.readLine());
    }
    else if (choice == 2) {
        System.out.println(in.readLine());
        double amt = sc.nextDouble();
        sc.nextLine();
        out.println(amt);

        System.out.println(in.readLine());
        System.out.println(in.readLine());
    }
}
```

```
else if (choice == 3) {  
    System.out.println(in.readLine());  
    session = false;  
}  
else {  
    System.out.println(in.readLine());  
}  
}  
}  
  
} catch (Exception e) {  
    e.printStackTrace();  
}  
}  
}
```